Normal Distribution

(A special case of Probability-Distributions)

Review

1. Data types

1.1 Discrete : Gender, Marital Status, Education-Level, Satisfaction, etc.

1.2 Continuous : Heigth, Weigth, Length, etc.

2. Sample description

2.1 Central tendency (e.g., *M*, *Mdn*) summarize the center of a data distribution.

2.2 Dispersion (e.g., *SD*, IQR) describe the spread or variability of the data.

Today's Topic

1. Probability VS. Empirical Probability

1.1 $P(A) \in [0, 1]$

1.2 $P(x > a)$, $P(x < b)$, $P(a \leq x < b)$

Note : $P(x = a) = 0$ ($x \sim$ Continuous)

2. Probability Dristribution

2.1 Discrete Data

Table 1

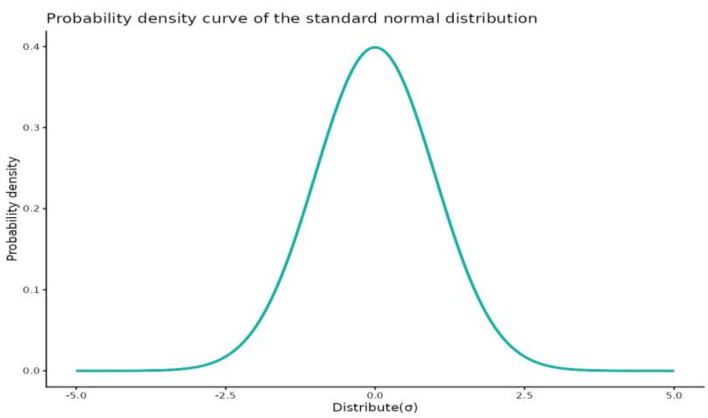
Probability Distribution of a Discrete Random Variable

X	≤6	7	8	9	10
p	0.15	0.30	0.35	0.15	0.05

Note. Data represent shooting scores.

2.2 Continuous Data

$$P(x < a) = \int_{-\infty}^{a} f(x)dx$$



2.3 Property of probability distribution

2.3.1 $p_i \geq 0$

2.3.2 $\sum_{i=1}^{n} p_i = 1$, x ~ Discrete; $\int_{-\infty}^{\infty} \mathrm{f(x)dx} = 1$, x ~ Continuous

3. Normal Distribution

3.1 Normal Distribution curve

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}, x \sim (\mu, \sigma^2)$$

3.2 Property of Normal Distribution curve

3.2.1 Unimodality: the curve reaches its maximum at the mean μ.

3.2.2 Symmetry: the curve is symmetric about the mean μ.

3.2.3 Two parameters: μ (location) and σ (scale, or spread). **(Appendix : R ShinyApp)**

3.2.4 Two inflection points: located at μ ± σ.

3.2.5 Domain: all real numbers, i.e., the curve extends from −∞ to +∞.

4. Standard Normal Distribution

4.1 Why Standard Normal (e.g. $)

4.1.1 Unified Reference: Transform any normal distribution to N(0,1)

4.1.2 Probability Calculation: One Z-table for all normal probabilities

4.1.3 Cross-Distribution Comparison: Z-scores enable comparison across different scales

4.1.4 Basis of Inferential Statistics: Critical values (CI) from Z-distribution

4.1.5 Outlier Detection: Objective rule: |Z| > 3 (3σ rule)

4.2 Transformation

$z = \frac{x-\mu}{\sigma}$; $(z = \frac{x-\bar{x}}{s})$ **(Appendix Derivation)**

4.3 Standard Normal Distribution curve

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, z \sim (0,1)$$

4.4 Property of Standard Normal Distribution curve

4.4.1 Unimodality: the curve reaches its maximum at the mean 0.

4.4.2 Symmetry: the curve is symmetric about the mean 0 (0.5 + 0.5).

4.4.3 Two parameters: 0 (location) and 1 (scale, or spread). **(Appendix : R ShinyApp)**

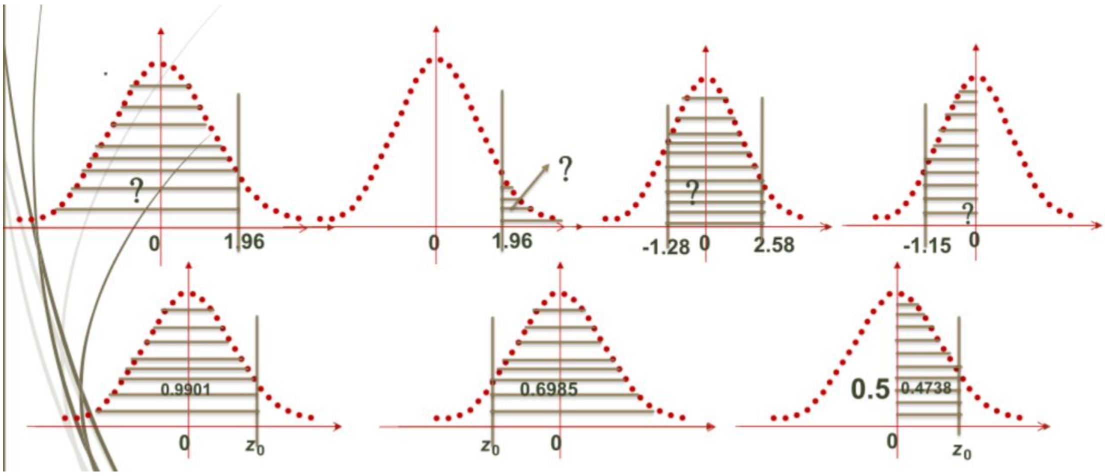
4.4.4 Two inflection points: located at ± 1.

4.4.5 Domain: all real numbers, i.e., the curve extends from $-\infty$ to $+\infty$.

5. Z-Table

5.1 From z to p

5.2 From p to z



Appendix Derivation

1. Normal-Distribution's Parameters

1.1 Data: $x_1, x_2, x_3, \ldots, x_n$;

1.2 Mean

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \ldots + x_n}{n}$$

1.3 Standard Deviation

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

2. Standard Normal-Distribution's Parameters

2.1 $z = \frac{x - \bar{x}}{s}$

2.2 Data: $\frac{x_1 - \bar{x}}{s}, \frac{x_2 - \bar{x}}{s}, \frac{x_3 - \bar{x}}{s}, \ldots, \frac{x_n - \bar{x}}{s}$

2.3 Mean

$$\frac{\frac{x_1 - \bar{x}}{s} + \frac{x_2 - \bar{x}}{s} + \frac{x_3 - \bar{x}}{s} + \ldots + \frac{x_n - \bar{x}}{s}}{n}$$

$$= \frac{1}{s}\left(\frac{x_1 + x_2 + x_3 + \ldots x_n - n\bar{x}}{n}\right) = \frac{1}{s}\left(\frac{x_1 + x_2 + x_3 + \ldots x_n}{n} - \frac{n\bar{x}}{n}\right)$$

$$= \frac{1}{s}(\bar{x} - \bar{x}) = 0$$

2.4 Standard Deviation

$$\sqrt{\frac{\left(\frac{x_1 - \bar{x}}{s}\right)^2 + \left(\frac{x_2 - \bar{x}}{s}\right)^2 + \left(\frac{x_3 - \bar{x}}{s}\right)^2 + \ldots + \left(\frac{x_n - \bar{x}}{s}\right)^2}{n-1}}$$

$$= \sqrt{\frac{\frac{1}{s^2}\left[(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + (x_3 - \bar{x})^2 + \ldots + (x_n - \bar{x})^2\right]}{n-1}} = \frac{1}{s}\sqrt{\frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + (x_3 - \bar{x})^2 + \ldots + (x_n - \bar{x})^2}{n-1}}$$

$$= \frac{1}{s} \times \frac{1}{s} = 1$$

Appendix

1. 概念与理论类

1.1 简述正态分布的特征

1.2 简述标准正态分布与一般正态分布的区别与联系

1.3 名词解释：正态性检验

2. 应用与计算类

2.1 百分等级与标准分数计算

例题：某体院运动心理学测试中，学生焦虑量表得分呈正态分布，平均分为 50，标准差为 10。请问：

(1)某生得分为 70 分，其标准分数是多少？在该群体中大约超过百分之多少的人？

(2)如果要选拔得分在前 10%的学生进行干预，分数线应划在多少分？

3. 易错点

3.1 “正态分布”：画钟形曲线并列出特征（对称、单峰、面积）

3.2 “正态性检验”：图示法（直方图、Q-Q 图）与统计检验法（Shapiro-Wilk 检验，适用于小样本；Kolmogorov-Smirnov 检验，适用于大样本）

3.3 公式：Z 分数计算